

EE263 Practice Midterm
Summer 2026

1. The Gram matrix $A^T A$. Let $A \in \mathbb{R}^{m \times n}$ be an arbitrary matrix. Answer the following questions about $A^T A$.

a) If $A^T A = 0$, then $A = 0$.

☐ True

☐ False

b) If $A^T A$ is upper triangular, then it must be diagonal.

☐ True

☐ False

c) If $m = n$ and $A^T A$ is diagonal, then A is an orthogonal matrix.

☐ True

☐ False

d) If A is square, then it is not possible for all the off-diagonal entries of $A^T A$ to be negative simultaneously.

☐ True

☐ False

2. A tall matrix with $U^T U = I$. If U is an $m \times n$ matrix such that $U^T U = I$ but $U U^T \neq I$, then what can we say about the set $S = \{x \mid U U^T x = x\}$?

☐ $S \subset \text{range}(U)$

☐ $\text{range}(U) \subset S$

☐ $S = \text{range}(U)$

☐ There is no particular relationship between S and $\text{range}(U)$.

3. Union of two subspaces. If S and T are two subspaces in $\mathbb{R}^n$ such that $0 < \dim(S) < \dim(T) < n$, then $S \cup T$:

☐ is never a subspace

☐ is a subspace only if $S \cap T = S$

☐ can be a subspace such that $\dim(S \cup T) > \dim(T)$

☐ is always a subspace

4. Nullspace of a product. If $\text{null}(A) = \{0\}$ and $\text{null}(B) = \{0\}$, then $\text{null}(AB) = \{0\}$.

- ☐ True
- ☐ False

5. Intersection of nullspaces. If $\text{null}(A) \cap \text{null}(B) = \{0\}$, then $\text{null}(AB) = \{0\}$.

- ☐ True
- ☐ False

6. Additivity and linearity. If for a function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ we have $f(x + y) = f(x) + f(y)$, then f can be represented as $f(x) = Ax$, where $A \in \mathbb{R}^{m \times n}$.

- ☐ True
- ☐ False

7. Powers of an orthogonal matrix. For any given $n \geq 2$, there exists a 2×2 orthogonal matrix U such that $U^n = I$ but $U^k \neq I$ for all $k < n$.

- ☐ True
- ☐ False

8. A recursive matrix construction. Consider the set of matrices constructed as follows:

$$H_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad H_n = \begin{bmatrix} H_{n-1} & H_{n-1} \\ H_{n-1} & -H_{n-1} \end{bmatrix} \quad (\text{for } n > 2).$$

These matrices are orthogonal.

- ☐ True
- ☐ False

9. A rank-one matrix. Suppose $A = uv^T$, where $u, v \in \mathbb{R}^n$ and $n \geq 3$.

a) A cannot be upper triangular.

- ☐ True
- ☐ False

b) The sum of the diagonal entries of A is equal to the inner product of u and v .

- ☐ True
- ☐ False

c) If $A = QR$ is the QR factorization of A obtained through the classical Gram-Schmidt procedure, which of the following statements is true?

- ☐ $Q = u$
- ☐ Q and u are parallel and point in the same direction
- ☐ Q and u are parallel but may point in opposite directions
- ☐ Q and u may or may not be parallel

10. Spanning a hyperplane. If $v_1, v_2, \dots, v_{n-1} \in \mathbb{R}^n$ are $n - 1$ linearly independent vectors, their linear combinations span a hyperplane.

- ☐ True
☐ False

11. A set defined by a distance inequality. Let $S = \{x \in \mathbb{R}^n \mid \|x - b\| \leq \|x\|\}$, where b is a vector in $\mathbb{R}^n$. Then S has a basis, *i.e.*, a set of independent vectors $V = \{v_1, \dots, v_k\}$ exists such that $S = \text{span}(V)$.

- ☐ True
☐ False

12. A wide system of equations. If $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, and $x \in \mathbb{R}^n$, such that $m < n$, then the linear system of equations $Ax = b$ has either no solution or infinitely many solutions.

- ☐ True
☐ False

13. Logical implications. In each part of this question, you will see two statements A and B . For each part, select the one choice that describes their logical relationship, where:

- $A \Rightarrow B$ means that if A is true, then B must also be true, but there are cases where B is true and A is not;
- $B \Rightarrow A$ means that if B is true, then A must also be true, but there are cases where A is true and B is not;
- $A \Leftrightarrow B$ means that the two statements are logically equivalent: each is true if and only if the other is true;
- $A \not\Rightarrow B$ means that neither statement implies the other.

a) Suppose $A, B \in \mathbb{R}^{m \times n}$ are two matrices.

(A) $\text{range}(A) \subseteq \text{range}(B)$.

(B) $\text{rank}(\begin{bmatrix} A & B \end{bmatrix}) = \text{rank}(B)$.

- ☐ $A \Rightarrow B$
☐ $B \Rightarrow A$
☐ $A \Leftrightarrow B$
☐ $A \not\Rightarrow B$

b) Let $A \in \mathbb{R}^{m \times n}$ and $y \in \mathbb{R}^m$.

(A) $y \notin \text{range}(A)$.

(B) $y \in \text{range}(A)^\perp$.

- ☐ $A \Rightarrow B$
☐ $B \Rightarrow A$

- ☐ $A \Leftrightarrow B$
- ☐ $A \not\Leftrightarrow B$
- c) Let $A \in \mathbb{R}^{m \times n}$, $x \in \mathbb{R}^n$, $y \in \mathbb{R}^m$, and $m > n$.
 - (A) Given y and A , the system of equations $y = Ax$ has infinitely many solutions.
 - (B) $\text{null}(A) \neq \{0\}$.

- ☐ $A \Rightarrow B$
- ☐ $B \Rightarrow A$
- ☐ $A \Leftrightarrow B$
- ☐ $A \not\Leftrightarrow B$

14. An operator on linear maps. For any arbitrary function like h , we define the operator $\mathcal{T}$, such that $\mathcal{T}(h)$ is the minimum number of vectors needed to span the subspace that is created by all the possible outputs of the function h . Let f and g be two functions from $\mathbb{R}^n$ to $\mathbb{R}^m$, such that for any given $x, y \in \mathbb{R}^n$, $c \in \mathbb{R}$, we have that $f(x + cy) = f(x) + c f(y)$ and $g(x + cy) = g(x) + c g(y)$.

- a) Let $m = n$. Then, $\mathcal{T}(f(g)) = \mathcal{T}(g(f))$.
 - ☐ True
 - ☐ False
- b) Let $m = n$. Then, $\mathcal{T}(f(f)) \leq \mathcal{T}(f) = \mathcal{T}(f + f)$.
 - ☐ True
 - ☐ False
- c) Let $m = n$. Then, while this might not be true for every f and g , we can find a specific pair of f, g such that $\mathcal{T}(f(g)) > \mathcal{T}(g)$.
 - ☐ True
 - ☐ False
- d) Let $m = n$. Then, while this might not be true for every g , we can find a specific g such that $\mathcal{T}(g) > 0$, and for any arbitrary f , we have that $\mathcal{T}(f(g)) = \mathcal{T}(g(f)) = \mathcal{T}(f)$.
 - ☐ True
 - ☐ False
- e) Let $m = n$. Then, while this might not be true for every g , we can find a specific g such that $\mathcal{T}(g) > 0$, and for any arbitrary f , we have that $\mathcal{T}(f(g)) = \mathcal{T}(g(f)) = \mathcal{T}(g)$.
 - ☐ True
 - ☐ False

f) Let $m = n$. Then, if $\mathcal{T}(f) > 0$, then $\mathcal{T}(f(f))$ is also positive.

☐ True

☐ False

g) Let $m = n$, and suppose we select f such that for any arbitrary g , we have that $\mathcal{T}(f) \geq \mathcal{T}(g)$. Then, $\mathcal{T}(f(f)) = \mathcal{T}(f)$.

☐ True

☐ False

h) Let $m > n$, and suppose we select f such that for any arbitrary g , we have that $\mathcal{T}(f) \geq \mathcal{T}(g)$. Then, a function h exists, such that for any $x \in \mathbb{R}^n$, we have that $h(f(x)) = x$.

☐ True

☐ False

i) Let $m > n$, and suppose we select f such that for any arbitrary g , we have that $\mathcal{T}(f) \geq \mathcal{T}(g)$. Then, a function h exists, such that for any $x \in \mathbb{R}^n$, we have that $f(h(x)) = x$.

☐ True

☐ False

j) Suppose we select f such that for any arbitrary g , we have $\mathcal{T}(f) \leq \mathcal{T}(g)$. Then, we can find an $x \in \mathbb{R}^n$ such that $\|f(x)\| > \|x\|$.

☐ True

☐ False